HAPPIEST HEALTH · DIGITAL TWIN CHALLENGE 2026

HEATSHIELD

A medication-aware digital twin that forecasts heat strain 24 hours ahead for older patients with hypertension, diabetes and kidney disease

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Prototype on synthetic data · Not for clinical use

THE PROBLEM

Indian heatwaves hit hardest in older patients on everyday heart and diabetes medicines

DIURETICS

Extra fluid loss

The patient starts the day with less water and dehydrates faster

SGLT2 INHIBITORS

Water follows glucose

Glucose passed into urine pulls water with it, adding to fluid loss

BETA-BLOCKERS

The warning sign goes quiet

Heart rate stops rising with body heat, so strain stays hidden

THE BLIND SPOT

Wearables guess core temperature from heart rate. On a beta-blocker, that guess fails.

THE STANDARD METHOD ASSUMES

Hot body → fast heart. Read the heart, infer the heat.

ON A BETA-BLOCKER

The body heats up but the heart stays calm, so strain is underestimated.

THE GAP

Existing heat-strain twins are built for healthy workers, soldiers and athletes.

WHAT HEATSHIELD IS

A virtual copy of each patient that learns, forecasts and answers “what if?”

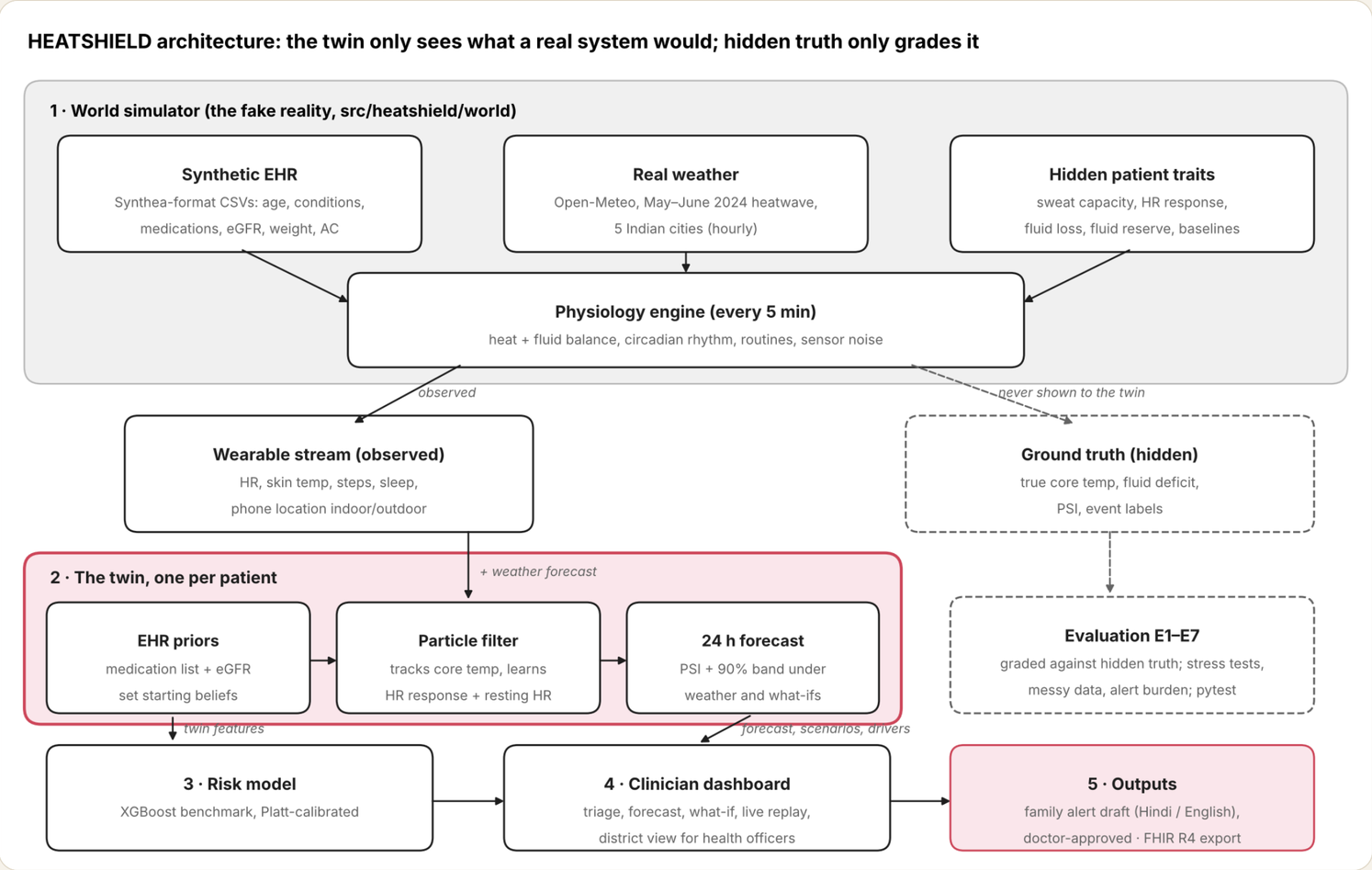
	Typical ML risk model	HEATSHIELD twin
Learns	Patterns across a population	**This patient’s own heat response**
Output	A risk score now	**A 24-hour forecast with an uncertainty band**
Updates	When retrained	**Every 5 minutes, with each reading**
Answers “what if?”	No	**Yes: re-simulates the patient**
Needs labelled outcomes	Yes	**No: the forecast is physiological**

ARCHITECTURE

The twin only sees what a real system would

A separate, richer world simulator plays reality. Its hidden traits and ground truth never reach the twin; they are used only to grade it.

This avoids the classic trap of a model reading its own equations back.



DATA: SYNTHETIC AND OPEN ONLY

Three streams, no real patient data

Health records

240 adults aged 55–85, in Synthea format: conditions, medications, eGFR, weight, plus home AC and outdoor routine

Wearable stream

Every 5 minutes for 8 days: heart rate, skin temperature, steps, sleep, with sensor noise and dropouts

Weather

Open-Meteo hourly history, May–June 2024 heatwave, in Jaipur, Delhi, Jodhpur, Nagpur and Lucknow

Every physiological and drug assumption lives in one file, **config/assumptions.yaml**, and is labelled illustrative.

HOW THE TWIN WORKS

Four steps, every five minutes

1 EHR priors

Medicines and eGFR set starting beliefs: beta-blocker → weak heart-rate response

2 Particle filter

400 particles track core temperature and fluid deficit, and learn this patient’s heart-rate response

3 24-hour forecast

Every particle runs forward under the weather forecast: a strain curve, a 90% band, P(high strain)

4 What-if and why

Change an input and re-run; rank drivers by how much risk drops when each is removed

RESULT 1 · MEDICATION-AWARENESS MATTERS

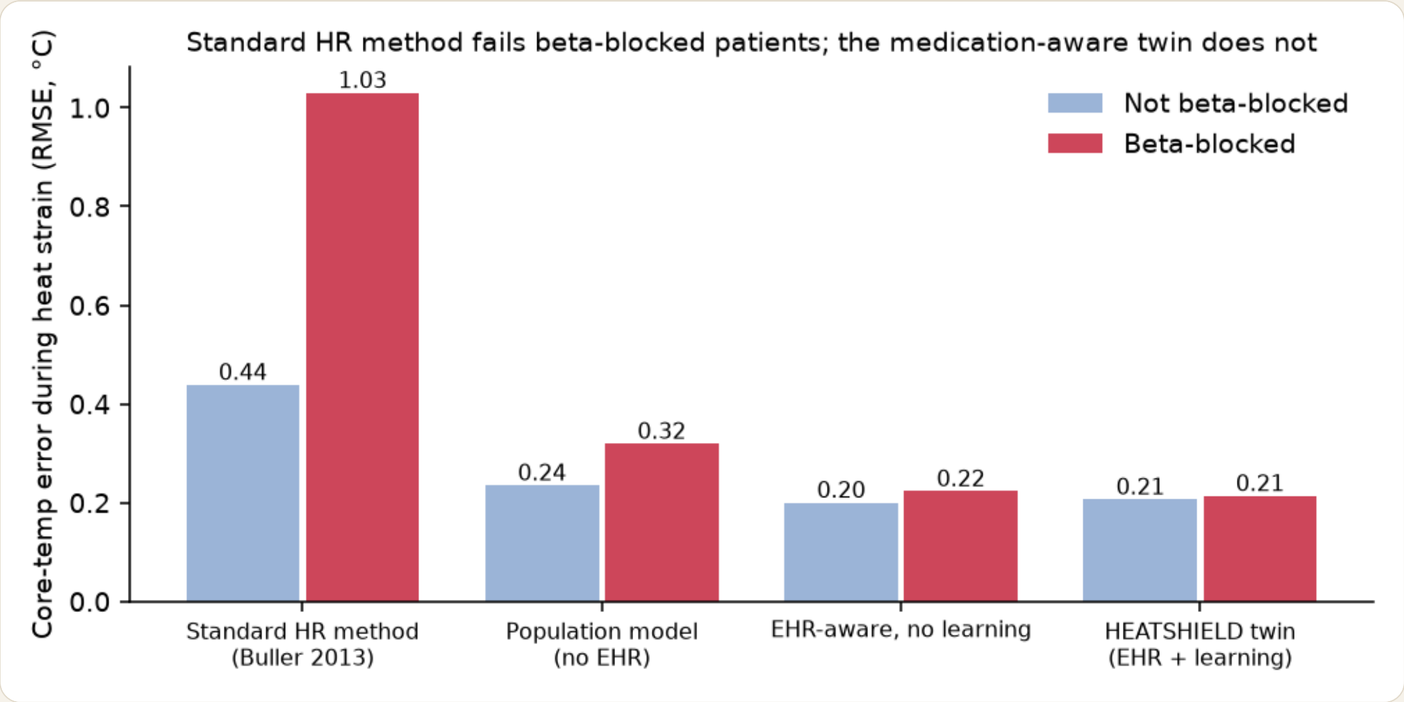
The standard method is 1 °C off for beta-blocked patients; the twin is not

1.03 °C

core-temperature error, standard HR method, beta-blocked patients under heat strain

0.21 °C

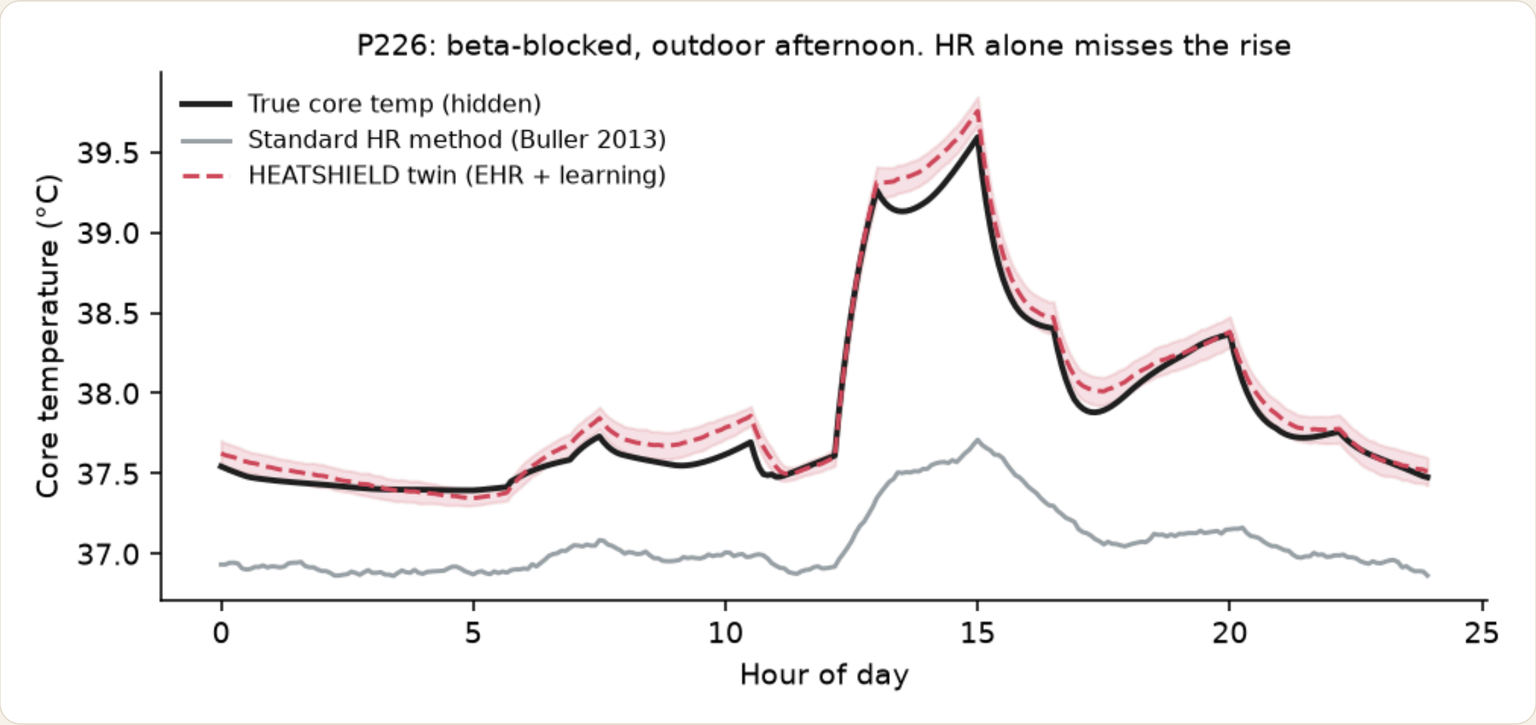
HEATSHIELD twin, the same patients, and the same for everyone else



RMSE when true core ≥ 37.8 °C · all 240 synthetic patients · real May–June 2024 heatwave weather

ONE PATIENT, ONE HOT AFTERNOON

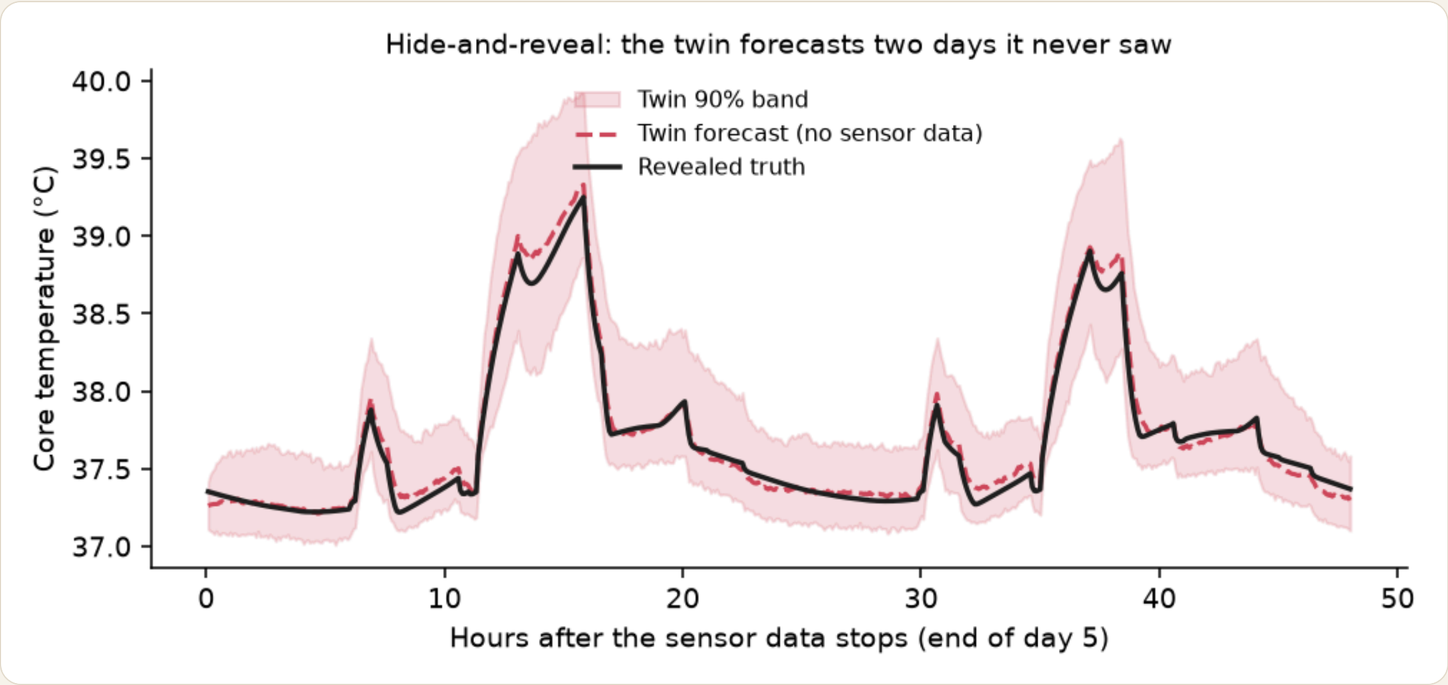
Heart rate alone reads him 2 °C too cool; the twin follows the rise



Synthetic patient P226, 85, beta-blocked, outdoor afternoon routine · true core temperature is hidden from the twin

RESULTS 2 AND 3 · LEARNING AND FORECASTING

Calibrate on five days, then forecast two it never saw



0.97

correlation between learned and hidden heart-rate response

0.29

peak-strain error, vs 0.90 for a population model

82%

of revealed truth inside the 90% band

Peak-strain error on the 0–10 Physiological Strain Index · all 240 synthetic patients

RESULT 4 · PREDICTING TOMORROW'S HIGH STRAIN

The twin alone matches a supervised model, without any labelled outcomes

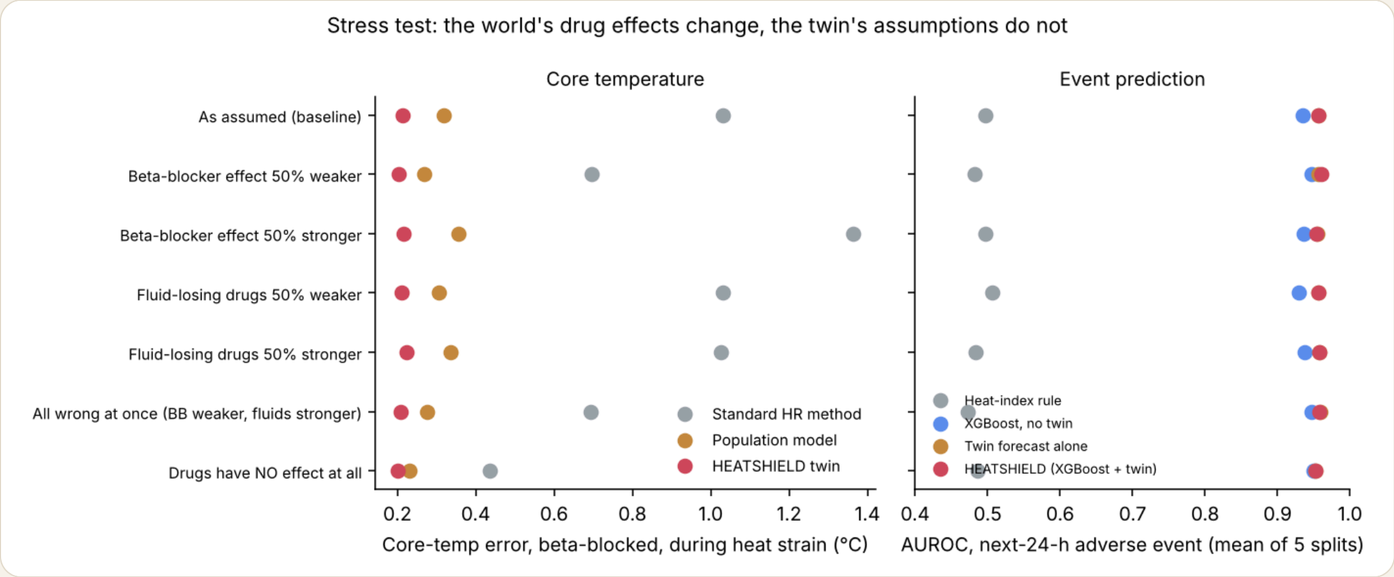
Model	AUROC	AUPRC	Brier
Heat-index rule (today's heat warnings)	0.50	0.10	0.59
XGBoost on EHR + wearable + weather	0.94	0.52	0.05
Twin forecast alone (no labels needed)	**0.96**	**0.59**	0.06
XGBoost + twin features	**0.96**	0.58	**0.05**

A one-size-fits-all heat warning does no better than chance at telling which patient will be strained. The twin's forecast needs no outcome labels, so it drives the dashboard; XGBoost is a benchmark and second opinion.

1,680 patient-evenings, 7.6% events · mean over 5 random patient-level splits · higher AUROC/AUPRC and lower Brier are better

RESULT 5 · STRESS TEST

Even when our drug assumptions are 50% wrong, the twin stays within 0.22 °C



0.20–0.22 °C

twin error in every world, because it learns each patient’s heart-rate response

0.28 °C

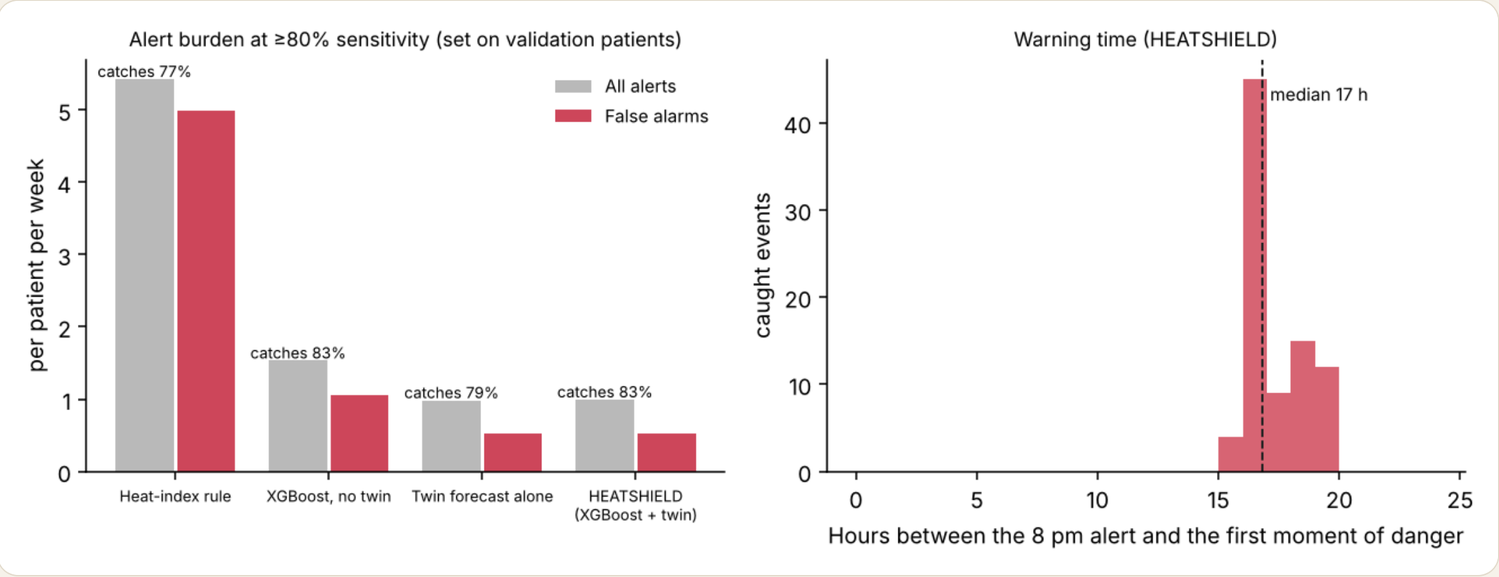
the same model without learning, when the beta-blocker effect is stronger

Honest finding: if the drugs had no effect at all, the twin’s edge over XGBoost disappears (AUROC 0.95 vs 0.95). Medication-awareness pays off only when the drug effect is real.

The simulator’s drug effects change; the twin keeps its original assumptions · experiments/robustness.py

RESULTS 6 AND 7 · MESSY DATA, ALERT BURDEN, WARNING TIME

Half the false alarms of XGBoost, with 17 hours of warning



1.0 / week

alerts per patient, 48% of them right

17 h

median warning: alert at 8 pm, danger the next afternoon

47% missing

watch data, and the twin's error only moves 0.21 → 0.22 °C

Thresholds set on validation patients to catch ~80% of events; heat-index rule: 5.4 alerts a week, only 8% right.

Test patients, mean of 5 splits · missing data: charging gaps, nights off-wrist, dropouts, motion spikes

THE CLINICIAN DASHBOARD

Each evening: who is at risk tomorrow, when, and why

Triage list

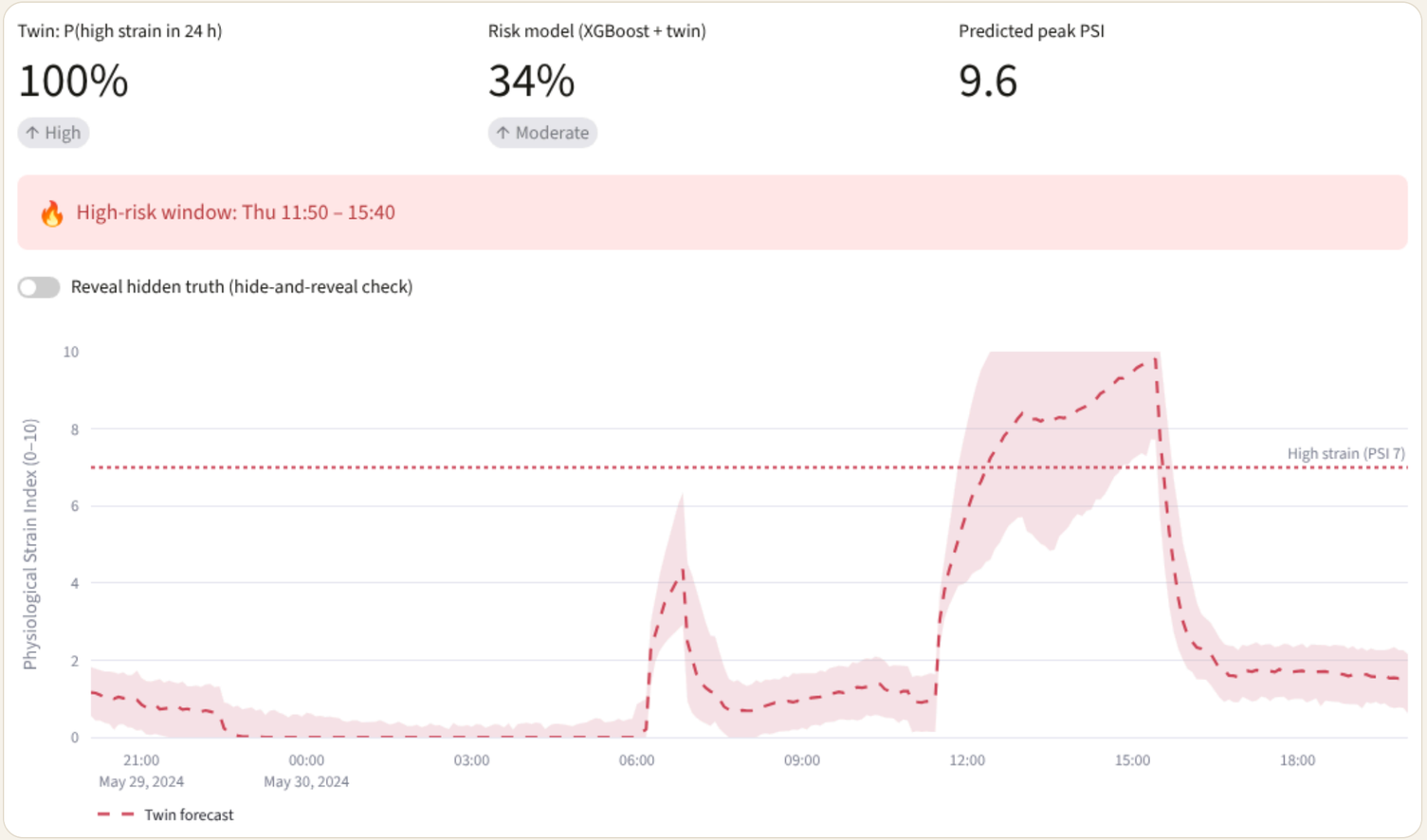
Patients ranked by tomorrow's risk, with heat-relevant medicines flagged

Forecast and window

A 24-hour strain curve with its band and the hours of highest risk

Main drivers

Outdoor routine, heat, AC, medicines, ranked by their share of the risk

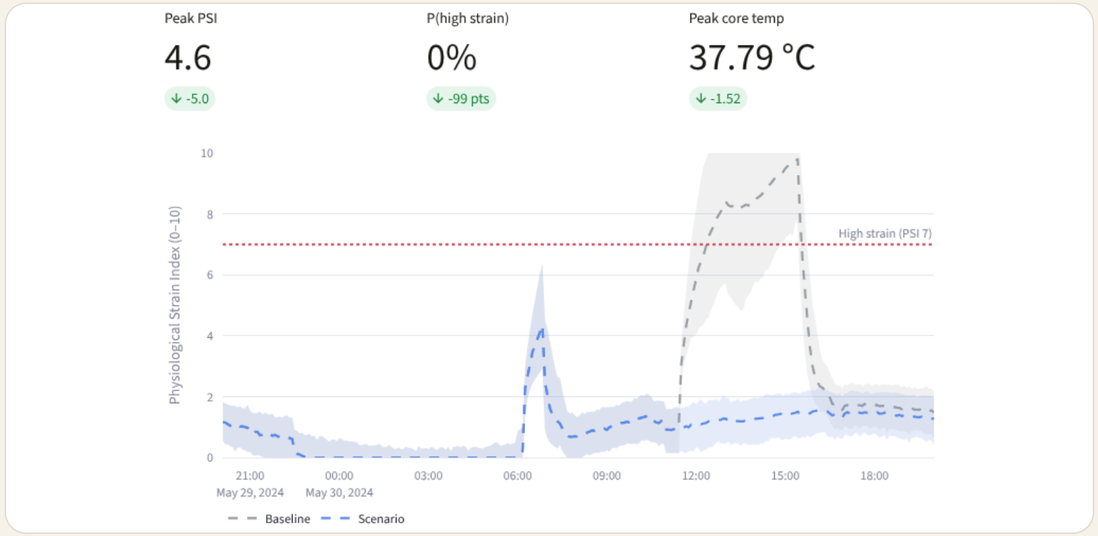


Streamlit + Plotly · solid = observed · dashed = estimated · shaded = uncertainty · dotted = hypothetical

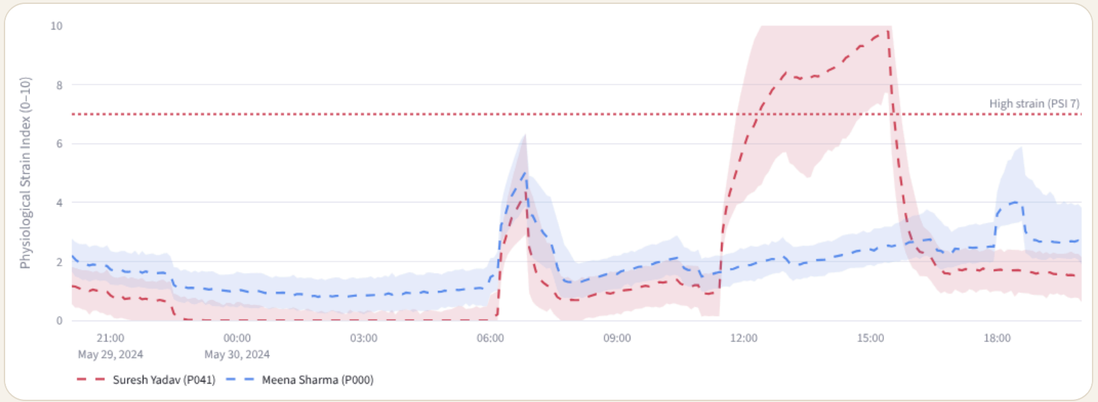
THE WHAT-IF SIMULATOR

Staying indoors 11 am–4 pm takes his risk from 99% to nearly zero

One patient, two scenarios. Peak strain 9.6 → 4.6, core temperature –1.5 °C



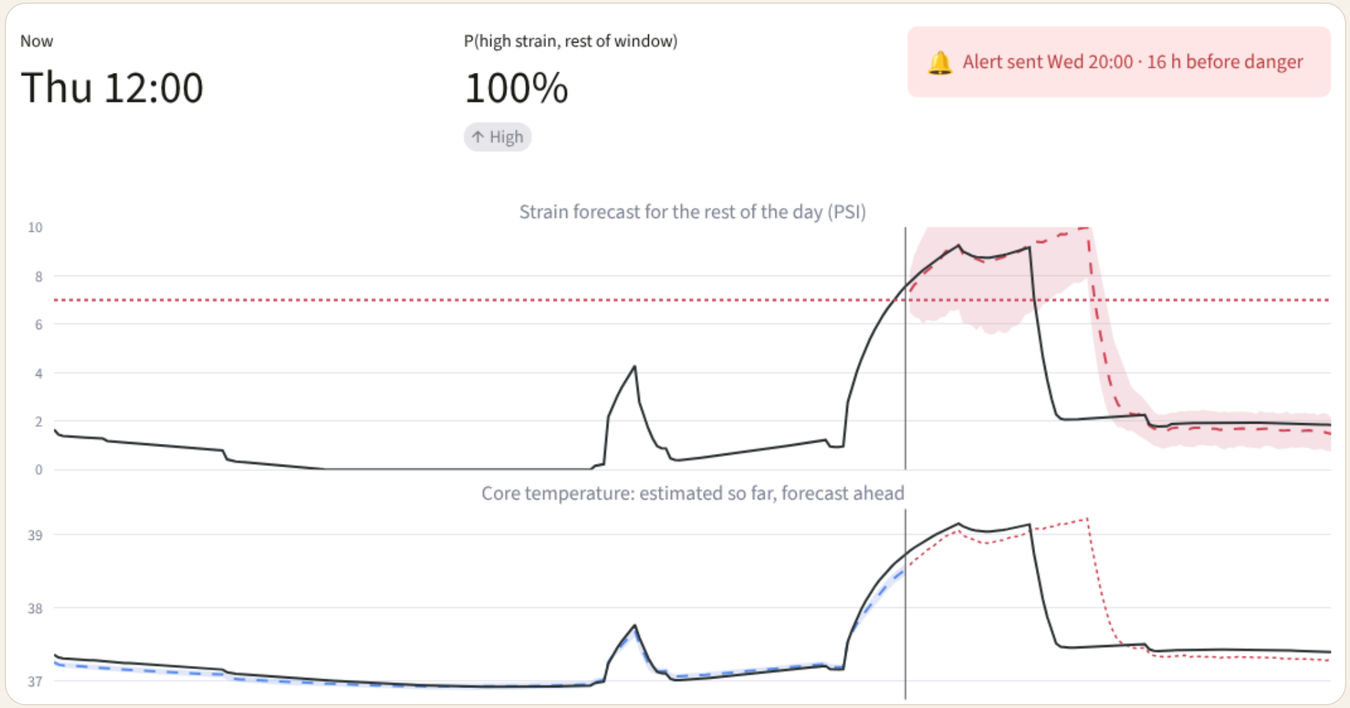
Two patients, same heatwave. Different medicines and routines, different peaks



Other levers: hotter day, power cut, lighter activity, extra fluids, and medication scenarios labelled **hypothetical, for clinical judgement only**.

LIVE REPLAY

Watch tomorrow unfold: the alert went out 16 hours before the danger



Every hour

the twin absorbs new watch readings and re-forecasts the rest of the day

Alert

sent the first time risk reaches 50%, here at 8 pm the evening before

Reveal

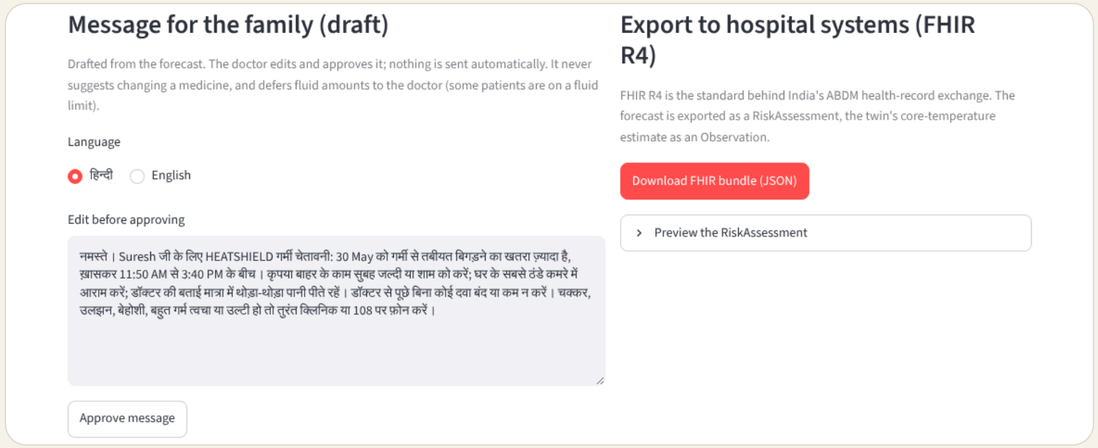
the hidden truth (black) is overlaid to show the forecast was right

Synthetic patient P041 · black = hidden truth, shown only for evaluation · dashed = twin

FROM FORECAST TO ACTION

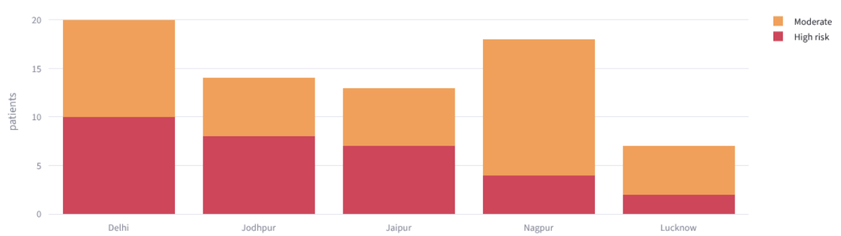
The family, the hospital record and the district team all get what they need

Family alert, Hindi or English. Drafted from the forecast, approved by the doctor



District view. High-risk patients by city, and how many have no AC

City	Patients	High risk	Moderate	High risk without AC	High risk on β-blocker	High risk with outdoor routine	Forecast max °C (mean)
Delhi	47	10	10	9	1	10	44.5
Jodhpur	52	8	6	7	6	8	44.2
Jaipur	43	7	6	4	2	7	44.5
Nagpur	49	4	14	2	1	4	43.8
Lucknow	49	2	5	1	1	2	42



FHIR R4 export: the forecast as a RiskAssessment and the core-temperature estimate as an Observation, the standard India’s ABDM health-record exchange is built on.

RESPONSIBLE USE

What HEATSHIELD claims, and what it does not

It does

- Show the method works in a simulated world under stated assumptions
- Stay accurate when its drug assumptions are 50% off, or 47% of watch data is missing
- Pass 12 automated checks: hotter never lowers risk, alerts never advise stopping a medicine
- Label every number observed, estimated, predicted or hypothetical

It does not

- Claim clinical accuracy: all data is synthetic, drug effects are illustrative
- Diagnose dehydration or kidney injury
- Recommend stopping or changing any medicine

Checks: tests/ · stress tests: experiments/robustness.py · assumptions: config/assumptions.yaml

PATH TO REAL PATIENTS

Swap the simulator for real streams; the twin stays the same

Connect

Smartwatch APIs, a FHIR feed into the twin (export is already built), live forecasts, an SMS gateway

Validate

A clinic pilot during a heatwave season, against heat-related visits, with ethics approval

Protect

Only data patients already share with their doctor, on the clinic’s own server, DPDP-compliant

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